Qt tools

The following are a few important tools provided by Qt SDK.

qmake is used to generate a platform-specific build script (Makefile) from project configuration (*.pro*) files. It is also used to create *.pro* files by inspecting available sources, headers and other files. Once the Makefile is generated, the *make* command can be used to build an application.

Qt Designer is used to create UI forms on the WYSIWYG principle. It allows you to create three types of forms—Window, Simple Dialogs, and Custom Widgets for re-usability. Many widgets are shown on the left-side pane of the *Designer*. You can drag and drop widgets onto the form. Apart from designing forms, it also supports property editing, signal/slot mapping, tab ordering, layout management, preview forms, etc. But *Designer* is not helpful for custom coding.

Qt Assistant is used to refer to documentation with great comfort.

Qt Linguist is used to translate user interfaces (UI) into local languages.

Qt Creator: As mentioned earlier, Qt Designer won't allow direct coding—you need to implement custom slots in the header, source files manually using an external editor and then use *qmake* and *make* from a terminal to build a project. This is not comfortable at all, compared to other GUI builders. However, with Qt Creator, application development can be really simple—just create a project on-the-fly and use the embedded form editor for direct coding. The context menu of any widget in the embedded editor provides an option to *'Go to Slot'* to write custom slots.

How to install

Download *qt-sdk-linux-x86-opensource-???.bin* from *qt.nokia.com*. (Use *qt-sdk-linux-x86_64-opensource-???.bin* for 64-bit platforms.)

Make the bin file executable by using the following commands:

```
chmod u+x qt-sdk-linux-x86-opensource-???.bin
```

Run the set-up as follows:

```
./qt-sdk-linux-x86-opensource-???.bin
```

Finally, follow the step-by-step procedure to finish installation.

If you run the installation as the root user, the SDK will be installed in */opt//qtsdk-???* by default and the Qt framework in */opt/qtsdk-???/qt/*, which we will refer to as QTDIR from now onwards. If you install it as a normal user, your home directory is used instead of */opt*.

Run *qtdemo* located in *$QTDIR/bin* to see Qt examples and demos, and to test them.

Check if *qt-creator* is available in the *Applications→Programs* menu after installing the SDK. Otherwise run *qt-creator* located in the */opt/qtsdk-???/bin* directory manually.

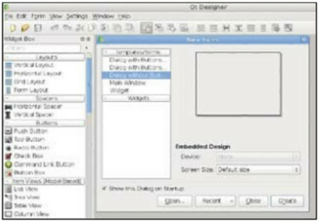

Figure 1: Qt Designer

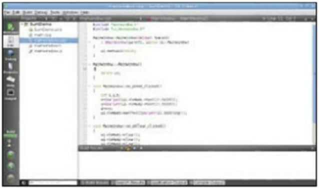

Figure 2: Qt Creator

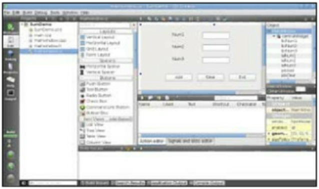

Figure 3: Editing form in the embedded editor of Qt Designer

Make sure the correct Qt framework is configured in *qt-creator* by navigating to *Tools→Options→Qt4→Qt versions*. Select the default option if more than one is available. At present, the latest stable releases are Qt Framework v4.5.2, Qt Creator v1.2.1.

Simple applications

In this example we will create a simple application to add two numbers.

Launch Qt Creator and create a new GUI-based project by navigating to *File→New→Projects→Qt4 Gui Application*. Key in a project name, the location, and select the modules to include. For this example no additional modules other than 'default' are required.